



TEST SCENARIO SPECIFICATION

Document ID	OC-REG-TS	Version	1.0
Project	OpenCart Platform	Module	User Registration
Techniques	EP · BVA · Decision Table	Status	Ready for Review
Total Scenarios	25 Test Scenarios	Prepared By	Mohamed Ahmed

SCENARIOS SUMMARY

Scenario ID	Title	TC	Category
TS-SUCC-001	Successful Account Registration	2	Functional
TS-NAME-001	Invalid First & Last Name	6	Validation
TS-NAME-002	First & Last Name Length Boundaries	5	Validation
TS-EMAIL-001	Invalid Email Formats	11	Validation
TS-EMAIL-002	Email Field Length Boundaries	6	Validation
TS-TEL-001	Invalid Telephone	6	Validation
TS-TEL-002	Telephone Field Length Boundaries	6	Validation
TS-TEL-003	Duplicate Telephone	1	Validation
TS-PASS-001	Password & Confirm Password Validation	8	Validation / Security
TS-PASS-002	Password Field Length Boundaries	6	Validation
TS-PP-001	Privacy Policy Validation	2	Functional
TS-NEWS-001	Newsletter Visibility & Default	1	Functional
TS-NEWS-002	Newsletter Mutual Exclusivity	1	Functional
TS-NEWS-003	Newsletter Preference Persistence	2	Functional
TS-NEWS-004	Default Newsletter Value Stored	1	Functional

TS-PERF-001	Registration Response Time	1	Performance
TS-SEC-001	SQL Injection Protection	1	Security
TS-SEC-002	XSS Attack Protection	1	Security
TS-SEC-003	Password Field Masking	1	Security
TS-USE-001	Clear Error Messaging	1	Usability
TS-USE-002	Password Visibility Toggle	1	Usability
TS-USE-003	Keyboard Navigation	1	Usability
TS-COMP-001	Responsive — Mobile (320–480px)	1	Compatibility
TS-COMP-002	Responsive — Tablet (768–1024px)	1	Compatibility
TS-COMP-003	Browser Compatibility (Chrome, Edge)	1	Compatibility

TEST SCENARIOS — DETAILED SPECIFICATIONS

1. Successful Account Registration

TS-SUCC-001	Validate Successful Account Registration
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<i>Given</i> the user is on the Registration page <i>When</i> the user enters <FName>, <LName>, <Email>, <Telephone>, <Pass>, and <Confirm> <i>And</i> the user sets Privacy Policy to <Privacy Policy> <i>And</i> clicks on the "Continue" button <i>Then</i> the registration result should be <Result>
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FName	LName	Email	Telephone	Pass	Confirm	Privacy Policy	Result
test	test	test@domain.com	012345	Test123	Test123	Accepted	Success
empty	empty	empty	empty	empty	empty	Not Accepted	Rejected

2. First & Last Name Validation

TS-NAME-001

Validate Registration is Rejected with Invalid First & Last Name

Given the user is on the Registration page
When the user enters <InvalidFirst & Last Name> in the First Name field
And fills in the rest of the required fields with valid data
And clicks on the " Continue " button
Then registration <result>

Case	Description / Invalid First & Last Name	Result
Missing	(empty)	Rejected
Numeric Characters	Mohamed123	Rejected
Special Characters	Mohamed@#\$	Rejected
Leading Spaces	Mohamed	Rejected
Trailing Spaces	Mohamed	Rejected
Only White Space	(spaces only)	Rejected

TS-NAME-002

Validate First And Last Name Length Boundaries

Given the user is on the Registration page
When the user enters a First or Last Name with <length> characters
And completes other fields with valid data, **And** accepts Privacy Policy, **And** clicks Continue
Then registration <result>

Case	Length	Result
Min	1	Success
Min + 1	2	Success
Max - 1	31	Success
Max	32	Success
Max + 1	33	Rejected

3. Email Validation

TS-EMAIL-001

Validate Registration is Blocked with Invalid Email Formats

Given the user is on the Registration page
When the user enters <InvalidEmail> in the Email field
And fills in the rest of the required fields
And clicks on the " Continue " button
Then registration <result>

Case	Description / Invalid Email	Result
Missing	(empty)	Rejected
Missing '@' Symbol	testdomain.com	Rejected
Missing Domain	test@	Rejected
Missing Username	@domain.com	Rejected
Leading Space	test@domain.com	Rejected
Trailing Space	test@domain.com	Rejected
Missing TLD	testuser@domain	Rejected
Double Dots	test..user@domain.com	Rejected
Disallowed Characters	te#st@domain.com	Rejected
Duplicate Email	test@domain.com	Rejected
Case-Insensitive	TEST@domain.com	Rejected

TS-EMAIL-002

Validate Email Field Length Boundaries

Given the user is on the Registration page
When the user enters Email with <length>
And fills in the rest of the required fields, **And** clicks on the " Continue " button
Then registration should be <Result>

Case Description	Length	Result
Min - 1	5	Rejected
Min	6	Accepted
Min + 1	7	Accepted
Max - 1	95	Accepted
Max	96	Accepted
Max + 1	97	Rejected

4. Telephone Validation

TS-TEL-001	Validate Registration is Rejected with Various Invalid Telephone
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Given the user is on the Registration page
When the user enters <InvalidTelephone> in Telephone field
And fills in the rest of the required fields with valid data
And clicks on the " Continue " button
Then registration <result>

Case	Description / Invalid Telephone	Result
Missing	(empty)	Rejected
Alphabetic Characters	test	Rejected
Special Characters	te#st	Rejected
Leading Space	1234	Rejected
Trailing Space	1234	Rejected
Only White Space	(spaces only)	Rejected

TS-TEL-002	Validate Telephone Field Length Boundaries
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Given the user is on the Registration page
When the user enters Telephone with <length>
And fills in the rest of the required fields, And clicks on the " Continue " button
Then registration should be <Result>

Case Description	Length	Result
Min - 1	2	Rejected
Min	3	Accepted
Min + 1	4	Accepted
Max - 1	31	Accepted
Max	32	Accepted
Max + 1	33	Rejected

TS-TEL-003	Validate Registration Fails for Duplicate Telephone
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Given the system has an existing account with Telephone <ExistingTelephone>
When the user enters <TelephoneNum> in the Telephone field
And fills in all other required registration fields, And clicks on the " Continue " button
Then registration should fail if it matches any existing telephone in the system.

5. Password Validation

TS-PASS-001	Validate Password and Confirm Password During Registration
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Given the user is on the Registration page
When the user enters <Password> in the Password field
And the user enters <ConfirmPassword> in the Confirm Password field
And fills in all other required registration fields
And clicks on the " Continue " button
Then registration should be <Result>

Case	Password	Confirm Password	Result
Empty	(empty)	(empty)	Rejected
Pass Missing	(empty)	test123	Rejected
Confirm Missing	test123	(empty)	Rejected
Not Matching	test123	test1234	Rejected
Case-Sensitive	TEST123	test123	Rejected
Leading Space	test123	test123	Rejected
Trailing Space	test123	test123	Rejected

Only White Space	(spaces)	(spaces)	Rejected
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TS-PASS-002	Validate Password Field Length Boundaries
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Given the user is on the Registration page When the user enters Password with <length> And fills in the rest of the required fields, And clicks on the "Continue" button Then registration should be <Result>
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Case Description	Length	Result
Min - 1	3	Rejected
Min	4	Accepted
Min + 1	5	Accepted
Max - 1	19	Accepted
Max	20	Accepted
Max + 1	21	Rejected

6. Privacy Policy Validation

TS-PP-001	Validate Privacy Policy Acceptance Business Rules
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Given the user is on the Registration page When the user fills <Mandatory> data in all required fields And the user sets Privacy Policy, And clicks "Continue" Then the registration result should be <Result>

Mandatory	Privacy Policy	Result
Missing	Accepted	Rejected
Valid	Not Accepted	Rejected

7. Newsletter Validation

TS-NEWS-001	Validate Newsletter Options Visibility and Default Selection
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Given the user is on the Registration page
Then the Newsletter section should display the following options: Yes / No
And only one option must be selectable at a time
And the default selection must follow defined business rules

TS-NEWS-002	Validate Mutual Exclusivity of Newsletter Options
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Given the user is on the Registration page
When the user selects the following options: Yes / No
Then <SelectedOption> should be selected
And the other option should be automatically deselected

TS-NEWS-003	Validate Newsletter Preference Persistence After Registration
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Given the user is on the Registration page
When the user selects <SelectedOption>
And the user completes all mandatory registration fields with valid data
And submits the registration form
Then the registration should be successful
And the newsletter preference must be stored correctly
And the stored value must match <SelectedOption>

TS-NEWS-004	Validate Default Newsletter Value is Stored When User Does Not Change Selection
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Given the user is on the Registration page
When the user completes registration without changing Newsletter option
And submits the registration form
Then the stored newsletter preference must match the default business rule value

8. Non-Functional Requirements Validation

Performance Validation

TS-PERF-001	Validate Registration Submission Response Time Under Normal Load
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Given the user is on the Registration page
And the user enters valid data in all mandatory fields
When the user clicks on "Register"
Then the registration process should complete successfully
And the total response time from click to confirmation message should be $\leq 3s$

Security Validation

TS-SEC-001	Validate Inputs are Protected Against SQL Injection Attacks
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Given the user is on the Registration page
When the user enters <SQLInjection> into <Field>, And submits the registration form
Then the system should reject the input
And no database error should be exposed
And a generic validation message should be displayed
And the application should remain stable

TS-SEC-002	Validate Inputs are Protected Against XSS Attacks
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When the user enters <XSSPayload> into <Field>, And submits the registration form
Then the input should be sanitized or rejected
And no script should execute
And no alert popup should appear
And the stored data should not contain executable script

TS-SEC-003	Validate Password Field is Masked by Default
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Given the user is on the Registration page
When the Registration page loads
Then the Password field input type should be "password"
And entered characters should be masked
And no plain text should be visible

Usability Validation

TS-USE-001

Validate Clear and User-Friendly Error Messaging

Given the user is on the Registration page
When the user submits invalid registration data
Then a clear, concise error message should appear
And the message should indicate which field is invalid
And the message should not reveal internal system logic
And error message should be positioned near the relevant field
And error text should be readable (contrast compliant)

TS-USE-002

Validate Password Visibility Toggle Availability & Consistency

Given the user is on the Registration page
When the page is loaded on any supported browser (Chrome, Edge)
Then a password visibility toggle control must be displayed next to the Password field
And the toggle must be implemented by the application (custom UI control)
And the toggle must not depend on browser-native password reveal features

TS-USE-003

Validate Keyboard Navigation Support

Given the user is on the Registration page
When the user navigates using keyboard only
Then pressing Tab should move focus forward logically
And pressing Shift + Tab should move focus backward
And pressing Enter on Submit should trigger form submission
And focus indicator should be visible on all interactive elements
And no keyboard trap should occur

Compatibility Validation

TS-COMP-001

Validate Registration Page is Fully Responsive on Mobile Devices

Given the user is on the Registration page
When the screen width is between 320px and 480px
Then all fields should be visible without horizontal scrolling
And layout should adapt to single-column structure
And buttons should be fully visible and tappable
And font size should remain readable without zoom

And no UI element should overlap
And error messages should wrap correctly

TS-COMP-002	Validate Registration Page is Fully Responsive on Tablet Devices
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Given the user is on the Registration page
When the screen width is between 768px and 1024px
Then the layout should adapt proportionally
And spacing between fields should remain consistent
And form elements should not stretch disproportionately
And touch targets should remain accessible
And orientation change (portrait ↔ landscape) should not break layout

TS-COMP-003	Validate Registration Page Compatibility on Major Browsers
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Given the user is on the Registration page
When the page is accessed via a supported browser
Then all UI elements should render correctly
And form validation should function consistently

DEFINITION OF DONE

✓	All Test Scenarios Executed and Passed
✓	No Critical / High Defects Open
✓	Cross-browser Compatibility Verified
✓	Performance Meets SLA (≤ 3s response time)
✓	Security Checks Passed (SQL Injection & XSS)